

LOW STYRENE SBR PATENT LANDSCAPE: EMULSION, SOLUTION, AND FUNCTIONALIZED STYRENE-BUTADIENE COPOLYMER ROUTES

1. ABSTRACT

This report synthesizes ten primary patents relevant to styrene-butadiene rubber (SBR) preparation, with emphasis on low-styrene embodiments and process routes that control styrene incorporation while preserving processability and tire performance. The selected patents span aqueous emulsion polymerization, organic-solvent solution polymerization, and post-polymerization functionalization/oil-extension strategies. Across the landscape, the target material is consistently a styrene-butadiene copolymer or elastomer, but the disclosed styrene levels vary widely: some patents explicitly use styrene in the 10–50 wt% range or around 25 wt%, while others disclose lower or moderate styrene fractions in solution SBR feeds (e.g., 30–40 wt% styrene in monomer charge) and functionalized emulsion SBRs with styrene around 21 wt% in the final polymer. Several patents focus less on lowering styrene per se and more on improving silica affinity, wet traction, rolling resistance, or processability through reactive emulsifiers, reactive polyols, epoxy-acrylate comonomers, silane modifiers, or oil extension. The dominant emulsion routes use water as the polymerization medium with rosin/fatty-acid emulsifier systems, redox or peroxide initiators, mercaptan chain-transfer agents, and low-temperature batch operation near 7–10 °C. The solution routes use hydrocarbon solvents such as hexane or cyclohexane, organolithium initiation, polar modifiers, and in some cases branching or coupling chemistry, followed by solvent removal and oil extension. Downstream water use is common in emulsion patents for coagulation, washing, and drying, while solution patents use water mainly for steam stripping or solvent removal. Overall, the patent set shows that low-styrene SBR development is not limited to a single synthesis platform; rather, it is achieved through careful monomer feed design and by integrating functional comonomers or post-modification steps that compensate for reduced styrene content with improved filler interaction and tire-relevant performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8530017B2
- Patent Title: ... copolymer using reactive emulsifier and styrene-butadiene copolymer prepared by ...
- Assignee: Korea Kumho Petrochemical Co., Ltd.
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses radical emulsion polymerization of styrene and 1,3-butadiene in water using a reactive emulsifier that participates in the polymerization and remains in the copolymer. The process is aimed at producing styrene-butadiene copolymers with improved silica affinity and tire tread performance, including variants containing reactive emulsifier-derived functionality. - Water: 1600 mL — Example 1 - Potassium rosinate: 11 g — Example 1 - Fatty acid potassium salt: 18 g — Example 1 - Potassium chloride: 1.5 g — Example 1 - Styrene: 260 g — Example 1 - Methylcarboxylate methacrylate: 30 g — Example 1 - 1,3-Butadiene: 710 g — Example 1 - EDTA: 0.5 g — Example 1 - Sodium hydrosulfite: 0.3 g — Example 1 - Ferrous sulfate: 0.1 g — Example 1 - Sodium formaldehyde sulfonate: 0.4 g — Example 1 - n-Dodecyl mercaptan: 1.0 g — Example 1 - Methane hydroperoxide: 0.6 g — Example 1 - Reactor volume: 5 L — Example 1 - Polymerization time: 8 hours — Example 1 - Polymerization temperature: 10 °C — Example 1 - Diethylhydroxyamine: 1.0 g — Example 1 - Weight average molecular weight: 860,000 g/mol — Example 1 - Styrene monomer amount: 10-50 wt % — Claims - Butadiene monomer amount: 45-85 wt % — Claims - Polymerization temperature: 0-80 °C — Synthesis passages - Polymerization time: 1-48 hours — Synthesis passages - Weight average molecular weight: 100,000-3,000,000 g/mol — Synthesis passages - Medium & Water Role: Water is the polymerization medium in the emulsion SBR synthesis, and later water is also used in downstream coagulation/stripping and washing steps. The examples explicitly charge water to the reactor and later describe stripping and drying after polymerization. - low styrene content: Styrene monomer amount: 10-50 wt % — Claims

Medium & Water Role - Water is the polymerization medium in the emulsion SBR synthesis, and later water is also used in downstream coagulation/stripping and washing steps. The examples explicitly charge water to the reactor and later describe stripping and drying after polymerization. - Core reaction medium: Aqueous emulsion / latex polymerization

low styrene content - Styrene monomer amount: 10-50 wt % — Claims

Relevant Experimental Evidence - Example 1: Reactive emulsifier-containing SBR was prepared with 260 g styrene and 710 g butadiene, giving Mw 860,000 g/mol. - Example 2: Increasing styrene to 280 g with polyoxyethylene phosphate methacrylate gave Mw 820,000 g/mol. - Example 3: Higher reactive emulsifier loading (30 g) still produced a high-Mw SBR (824,000 g/mol). -

Example 4: Allyl alcohol ethoxylate comonomer yielded a copolymer with Mw 850,000 g/mol. -

Example 7: Polyoxypropylene glycol phosphate acrylate comonomer produced Mw 845,000 g/mol.

Technical Relevance This patent is directly relevant because it is an SBR emulsion polymerization platform and explicitly discloses styrene/butadiene feed ratios in the low-to-moderate SBR range. It also shows how reactive emulsifiers can be used to tune silica affinity without abandoning the SBR backbone.

Patent 2

- Patent Number: US6858683B2
- Patent Title: Process for the preparation of SBR rubbers with an improved processability and ...
- Assignee: Polimeri Europa S.P.A.
- Jurisdiction: US
- Publication Year: 2005
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses anionic solution polymerization of styrene and butadiene in hydrocarbon solvent, followed by termination, oil extension, solvent stripping, and drying. It focuses on improving processability and rolling resistance through control of microstructure, branching, and reactor configuration rather than emulsion chemistry. - Cyclohexane: 15 liters (11.7 kg) — Comparative Example 1 - Tetrahydrofurfuryl ethyl ether (THFA): 2.4 g (200 ppm) — Comparative Example 1 - Butadiene: 1.14 kg — Comparative Example 1 - Styrene: 0.38 kg — Comparative Example 1 - Temperature: 30 °C — Comparative Example 1 - Li n-butyl: 2.05 g of a 5 wt% solution — Comparative Example 1 - Reaction time: 20 minutes — Comparative Example 1 - Temperature: 80 °C — Comparative Example 1 - Ethyl alcohol: 0.15 g — Comparative Example 1 - Highly aromatic oil: 0.560 kg — Comparative Example 1 - Water: present in solvent-removal vessel — Comparative Example 1 - Vapour temperature: 100 °C — Comparative Example 1 - Vacuum oven temperature: 60 °C — Comparative Example 1 - Drying time: 24 hours — Comparative Example 1 - CSTR volume: 100 liters each — Comparative Example 2 - Operating pressure: 3 bars — Comparative Example 2 - Residence time: 30 minutes — Comparative Example 2 - Butadiene: 9 wt % — Comparative Example 2 - Styrene: 3 wt % — Comparative Example 2 - Randomizing agent/vinyl promoter: 250 ppm — Comparative Example 2 - Temperature: 57 °C — Comparative Example 2 - Conversion: 74% — Comparative Example 2 - Temperature: 66 °C — Comparative Example 2 - Overall conversion: 99% — Comparative Example 2 - Vinyl unit content: 67% — Comparative Example 2 - Weight average molecular weight: 540,000 — Comparative Example 2 - Mooney viscosity: 51 — Comparative Example 2 - Cyclohexane: 15 liters (11.7 kg) — Comparative Example 3 - THFA: 4.9 g (400 ppm) — Comparative Example 3 - Butadiene: 1.14 kg — Comparative Example 3 - Styrene: 0.38 kg — Comparative Example 3 - Li n-butyl: 3 g of a 5 wt% solution — Comparative Example 3 - SiCl₄: 1.59 g of a 5 wt% solution in cyclohexane — Comparative Example 3 -

Temperature: 80 °C — Comparative Example 3 - Highly aromatic oil: 0.560 kg — Comparative Example 3 - Water: present in solvent-removal vessel — Comparative Example 3 - Vacuum oven temperature: 60 °C — Comparative Example 3 - Drying time: 24 hours — Comparative Example 3 - Weight average molecular weight: 649,000 — Comparative Example 3 - 1,2 content: 65% — Comparative Example 3 - Mooney viscosity: 49 — Comparative Example 3 - Residence time: 30 minutes — Comparative Example 4 - Butadiene: 9 wt % — Comparative Example 4 - Styrene: 3 wt % — Comparative Example 4 - Randomizing agent/vinyl promoter: 350 ppm — Comparative Example 4 - Temperature: 53 °C — Comparative Example 4 - Conversion: 59% — Comparative Example 4 - Temperature: 61 °C — Comparative Example 4 - Overall conversion: 90% — Comparative Example 4 - Temperature: 66 °C — Comparative Example 4 - Final conversion: 98% — Comparative Example 4 - SiCl₄: 40% of lithium fed — Comparative Example 4 - Highly aromatic oil: 27% — Comparative Example 4 - Weight average molecular weight: 720,000 — Comparative Example 4 - Vinyl unit content: 66% — Comparative Example 4 - Mooney viscosity: 50 — Comparative Example 4 - DVB: total quantity equal to lithium fed — Comparative Example 10 - Gel content: 5% — Comparative Example 10 - DVB: 75% to first reactor and 25% to second reactor — Comparative Example 11 - Insoluble product: 22% — Comparative Example 11 - Medium & Water Role: The polymerization itself is carried out in cyclohexane solution, not in water. Water appears later only in the solvent-removal step, where the polymer solution is fed to a stirred reactor containing water heated by steam to eliminate solvent, followed by filtration and drying. - low styrene content: Styrene: 3 wt % — Comparative Example 2; Styrene: 3 wt % — Comparative Example 4

Medium & Water Role - The polymerization itself is carried out in cyclohexane solution, not in water. Water appears later only in the solvent-removal step, where the polymer solution is fed to a stirred reactor containing water heated by steam to eliminate solvent, followed by filtration and drying. - Core reaction medium: Organic hydrocarbon solvent solution polymerization (cyclohexane)

low styrene content - Styrene: 3 wt % — Comparative Example 2; Styrene: 3 wt % — Comparative Example 4

Relevant Experimental Evidence - Comparative Example 1: Batch solution SBR was made in cyclohexane and finished by steam stripping and drying. - Comparative Example 2: Continuous solution SBR with 9 wt% butadiene and 3 wt% styrene gave 99% overall conversion and Mw 540,000. - Comparative Example 3: Radial batch SBR using SiCl₄ coupling produced Mw 649,000 and 1,2 content 65%. - Comparative Example 4: Continuous radial SBR with SiCl₄ coupling reached 98% final conversion and Mw 720,000. - Comparative Example 9: Branched continuous SBR using DVB showed no gel and Mw 650,000.

Technical Relevance This patent is relevant because it is a core SBR manufacturing disclosure, albeit in solution rather than emulsion form. It provides a low-styrene feed example in continuous operation and shows how reactor design and branching/coupling affect processability.

Patent 3

- Patent Number: US9328176B2
- Patent Title: Functional styrene-butadiene copolymer
- Assignee: Korea Kumho Petrochemical Co., Ltd.
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses emulsion polymerization of styrene-butadiene copolymers and functionalized variants in water, followed by ring-opening of epoxy groups using acidic or alkaline aqueous solutions. The process is designed to improve silica affinity and tire performance by introducing hydroxyl functionality into the SBR backbone. - Water: 1,500 mL — Example 1 - Sodium rosinate: 25 g — Example 1 - Sodium salt of fatty acid: 35 g — Example 1 - Styrene: 380 g — Example 1 - Glycidyl methyl methacrylate: 50 g — Example 1 - 1,3-Butadiene: 570 g — Example 1 - EDTA: 0.6 g — Example 1 - Ferrous sulfate: 0.1 g — Example 1 - n-Dodecylmercaptan: 1.5 g — Example 1 - p-Menthane hydroperoxide: 0.6 g — Example 1 - Reactor volume: 5 L — Example 1 - Temperature: 10 °C — Example 1 - Reaction time: 24 hours — Example 1 - Diethylhydroxylamine: 1.0 g — Example 1 - Sulfuric acid: 20% aqueous solution, 100 g — Example 1 - Average molecular weight: 908,000 g/mol — Example 1 - NaOH: 20% aqueous solution, 100 g — Example 4 - Average molecular weight: 740,000 g/mol — Example 4 - KOH: 20% aqueous solution, 100 g — Example 5 - Average molecular weight: 791,000 g/mol — Example 5 - Water: 1,500 mL — Comparative Example 1 - Styrene: 400 g — Comparative Example 1 - 1,3-Butadiene: 600 g — Comparative Example 1 - Average molecular weight: 843,000 g/mol — Comparative Example 1 - Water: 1,500 mL — Comparative Example 2 - Styrene: 380 g — Comparative Example 2 - 1,3-Butadiene: 570 g — Comparative Example 2 - Methyl methacrylic acid: 50 g — Comparative Example 2 - Average molecular weight: 795,000 g/mol — Comparative Example 2 - Styrene monomer amount: 10-50 wt % — Synthesis passages - Butadiene monomer amount: 45-85 wt % — Synthesis passages - Epoxy acrylate monomer amount: 0.1-10 wt % — Synthesis passages - Radical initiator amount: 0.05-3 parts by weight per 100 parts monomers — Synthesis passages - Polymerization temperature: 0-80 °C — Synthesis passages - Medium & Water Role: Water is part of the emulsion polymerization medium and is also used later in aqueous ring-opening chemistry and downstream latex handling. The examples explicitly charge water to the reactor and use 20% aqueous sulfuric acid, NaOH, or KOH for post-polymerization ring opening. - low styrene content: Styrene monomer amount: 10-50 wt % — Synthesis passages

Medium & Water Role - Water is part of the emulsion polymerization medium and is also used later in aqueous ring-opening chemistry and downstream latex handling. The examples explicitly charge water to the reactor and use 20% aqueous sulfuric acid, NaOH, or KOH for post-polymerization ring opening. - Core reaction medium: Aqueous emulsion / latex polymerization

low styrene content - Styrene monomer amount: 10-50 wt % — Synthesis passages

Relevant Experimental Evidence - Example 1: GMA-containing SBR was prepared in water and ring-opened with 20% sulfuric acid to give hydroxyl-functional copolymer. - Example 4: NaOH ring-opening produced a hydroxyl-functional SBR with Mw 740,000. - Example 5: KOH ring-opening produced a hydroxyl-functional SBR with Mw 791,000. - Comparative Example 1: Unfunctionalized SBR was prepared under similar emulsion conditions. - Comparative Example 2: Methyl methacrylic acid comonomer gave a different copolymer with Mw 795,000.

Technical Relevance This patent is relevant because it is an SBR emulsion process with explicit styrene/butadiene feed levels and a functional comonomer strategy that preserves SBR identity while improving silica compatibility. It is useful for low-styrene SBR research because it shows how monomer balance and post-ring-opening functionality can be combined in aqueous SBR synthesis.

Patent 4

- Patent Number: US8455600B2
- Patent Title: ... butadiene copolymer highly functionalized with reactive polyol and manufacturing ...
- Assignee: Korea Kumho Petrochemical Co., Ltd.
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses emulsion polymerization of styrene and butadiene in water using reactive polyol monomers, followed by preparation of functionalized SBR latex and downstream compounding. The reactive polyol is introduced during polymerization to improve silica affinity and blending behavior. - Water: 1600 mL — Example 1 - Potassium rosinate: 23 g — Example 1 - Fatty acid potassium salt: 37 g — Example 1 - Potassium chloride: 3.5 g — Example 1 - Styrene: 480 g — Example 1 - Reactive polyol (Chemical Formula 4): 30 g, molecular weight 400 — Example 1 - 1,3-Butadiene: 490 g — Example 1 - EDTA: 0.5 g — Example 1 - Sodium hydrosulfite: 0.3 g — Example 1 - Ferrous sulfate: 0.1 g — Example 1 - Sodium formaldehyde sulfonate: 0.4 g — Example 1 - Dodecanethiol: 1.0 g — Example 1 - Methane hydroperoxide: 0.6 g — Example 1 - Reactor volume: 5 L — Example 1 - Reaction time: 8 hours — Example 1 - Temperature: 10 °C — Example 1 - Diethylhydroxyamine: 1.0 g — Example 1 - Average molecular weight: 1,010,000 g/mol — Example 1 - Reactive polyol molecular weight: 3000 — Example 2 - Average molecular weight: 950,000 g/mol — Example 2 - Reactive polyol molecular weight: 9000 — Example 3 - Average molecular weight: 975,000 g/mol — Example 3 - Styrene: 450 g — Example 4 - Acrylonitrile: 30 g — Example 4 - Average molecular weight: 920,000 g/mol — Example 4 - Styrene: 430 g — Example 5 - Acrylonitrile: 50 g — Example 5 - Average molecular weight: 930,000 g/mol —

Example 5 - Bound styrene content: 21.0 wt% — Example 6 - cis-1,4 content: 9.5 wt% — Example 6 - trans-1,4 content: 76.3 wt% — Example 6 - vinyl 1,2 content: 14.2 wt% — Example 6 - Mooney viscosity (ML 1+4, 100°C): 45.8 — Example 6 - Average molecular weight: 356,000 g/mol — Example 6 - Medium & Water Role: Water is the polymerization medium for the SBR latex synthesis, and later water is also used in latex handling and compounding-related steps. The examples explicitly charge water to the reactor and describe the latex as the product form before oil blending and compounding. - low styrene content: Bound styrene content: 21.0 wt% — Example 6

Medium & Water Role - Water is the polymerization medium for the SBR latex synthesis, and later water is also used in latex handling and compounding-related steps. The examples explicitly charge water to the reactor and describe the latex as the product form before oil blending and compounding. - Core reaction medium: Aqueous emulsion / latex polymerization

low styrene content - Bound styrene content: 21.0 wt% — Example 6

Relevant Experimental Evidence - Example 1: Reactive polyol was incorporated during emulsion polymerization to give a styrene-butadiene-reactive polyol copolymer. - Example 2: Higher-molecular-weight reactive polyol was used, yielding Mw 950,000 g/mol. - Example 3: Sorbitol-derived reactive polyol gave Mw 975,000 g/mol. - Example 4: Acrylonitrile-containing functional copolymer was prepared with Mw 920,000 g/mol. - Example 6: Oil-extended functionalized SBR showed 21.0 wt% bound styrene and Mw 356,000 g/mol.

Technical Relevance This patent is relevant because it is an emulsion SBR platform with explicit styrene/butadiene feed levels and a functional monomer strategy that improves silica affinity. It also provides a concrete low-to-moderate styrene final polymer composition in a functionalized SBR embodiment.

Patent 5

- Patent Number: US20140256858A1
- Patent Title: Solution polymerization prepared styrene/butadiene elastomer containing liquid ...
- Assignee: Robert Anthony Bethea
- Jurisdiction: US
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses anionically initiated solution polymerization of styrene and 1,3-butadiene in hexane to make a high-molecular-weight SSBR cement, followed by blending with liquid styrene-butadiene polymer or petroleum oil and solvent removal. The focus is on extending the cement before recovery to improve processability. - Reactor volume: 2000 liters — Example I - Temperature: 60-70 °C —

Example I - Internal pressure: 350-450 kPa — Example I - Total monomers in hexane: 15 wt% — Example I - 1,3-Butadiene: 60 wt% of total monomers — Example I - Styrene: 40 wt% of total monomers — Example I - TMEDA: 0.12 ppm — Example I - n-BuLi: 0.039 ppm — Example I - Conversion: 90-99% — Example I - Polymerization termination agent: Polystay K and isopropyl alcohol, 1.0 and 0.5 ppm — Example I - Liquid styrene/butadiene polymer number average molecular weight: 1,000-50,000 g/mol — Claims - H-SSBR number average molecular weight: 200,000-1,000,000 g/mol — Claims - Triglyceride vegetable oil: 5-60 phr — Claims - Mooney viscosity: 50-180 — Claims - Mooney viscosity: 80-120 — Claims - Medium & Water Role: The polymerization is carried out in hexane solution, not in water. Water appears later only in steam-stripping/solvent-removal operations used to recover the oil-extended SSBR composite. - low styrene content: Not disclosed in extracted evidence

Medium & Water Role - The polymerization is carried out in hexane solution, not in water. Water appears later only in steam-stripping/solvent-removal operations used to recover the oil-extended SSBR composite. - Core reaction medium: Organic solvent solution polymerization (hexane cement)

low styrene content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example I: A hexane solution with 60 wt% butadiene and 40 wt% styrene was anionically polymerized to high conversion. - Example I: The recovered H-SSBR had bound styrene 38.9 wt% and Mooney viscosity about 121. - Example I: Petroleum oil extension reduced Mooney viscosity to about 52.8. - Example I: Soybean oil extension reduced Mooney viscosity further to about 40.

Technical Relevance This patent is relevant as a solution-polymerized SSBR route that explicitly controls styrene/butadiene feed ratio and demonstrates post-polymerization extension strategies. Although not a low-styrene patent, it is important for understanding solution SBR manufacturing and processability control.

Patent 6

- Patent Number: US10174133B2
- Patent Title: Modified styrene-butadiene copolymer, preparation method thereof, and rubber ...
- Assignee: Lg Chem, Ltd.
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - low styrene content: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence.

low styrene content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Comparative Example: Comparative Example - Example 1::
Example 1: - Example 2: Example 2 - Experimental Example 1: Experimental Example 1 -
Example 1: Example 1 - Example: Example

Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: EP2657262B1
- Patent Title: Triglyceride containing solution polymerization prepared styrene/butadiene ...
- Assignee: The Goodyear Tire & Rubber Company
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses anionically initiated solution polymerization of styrene and butadiene in hexane to form SSBR cement, followed by extension with triglyceride vegetable oil or petroleum oil and solvent removal. The examples compare petroleum oil and soybean oil extension effects on processability and Mooney viscosity. - Reactor volume: 2000 liters — Example I - Temperature: 63-71 °C — Example I - Internal pressure: 97-186 kPa — Example I - Total monomers in hexane: 12 wt% — Example I - 1,3-Butadiene: 70 wt% of total monomers — Example I - Styrene: 30 wt% of total monomers — Example I - TMEDA: 0.12 ppm — Example I - SMT: 0.0035 ppm — Example I - n-BuLi: 0.025 ppm — Example I - Conversion: 90-95% — Example I - Polystay K: 0.5 ppm — Example I - Bound styrene: 31.7 wt% — Example I - Vinyl content: 63.5% — Example I - cis content: 21.4% — Example I - trans content: 15.1% — Example I - Mooney viscosity: 107 — Example I - Soybean oil extension: 36.9 pphr — Example I - Petroleum oil extension: 36.8 pphr — Example I - Mooney viscosity after petroleum oil extension: 52.8 — Example I - Mooney viscosity after soybean oil extension: 40 — Example I - Triglyceride vegetable oil: 5-60 phr — Claims - H-SSBR Mooney viscosity: 50-180 — Claims - H-SSBR Mooney viscosity: 80-120 — Claims - Medium & Water Role: The polymerization is conducted in hexane solution, while water is used later in steam stripping to remove solvent from the SSBR/oil blend. No aqueous polymerization medium is disclosed. - low styrene content: Not disclosed in extracted evidence

Medium & Water Role - The polymerization is conducted in hexane solution, while water is used later in steam stripping to remove solvent from the SSBR/oil blend. No aqueous polymerization medium is disclosed. - Core reaction medium: Organic solvent solution polymerization (hexane cement)

low styrene content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example I: Base SSBR was prepared from 70 wt% butadiene and 30 wt% styrene in hexane. - Example I: The recovered SSBR had bound styrene 31.7 wt% and Mooney viscosity 107. - Example I: Petroleum oil extension reduced Mooney viscosity to 52.8. - Example I: Soybean oil extension reduced Mooney viscosity further to 40.

Technical Relevance This patent is relevant because it is a solution SBR process with a relatively lower styrene feed than many conventional grades and explicit control of polymer properties through oil extension. It is especially useful for understanding how solution SBR can be processed and extended without changing the SBR backbone chemistry.

Patent 8

- Patent Number: EP2424740B1
- Patent Title: Elastomer compositions modified by silanes
- Assignee: Dow Corning Corporation
- Jurisdiction: EP
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses modification of diene elastomers with olefinically unsaturated silanes in the absence of free-radical initiator, primarily as a compounding/modification route rather than SBR synthesis. The examples shown are natural rubber-based rubber compositions, so the patent is only indirectly relevant to SBR processing and silica coupling concepts. - Silica: 50.0 parts — Example 1 and 2 - Silane 1: 5.5 parts — Example 2 - Silane 2: 5.3 parts — Example 1 - Carbon black N234: 45.0 parts — Comparative example C1 - ACST: 2.5 parts — Example 1 and 2 - ZnO: 3.0 parts — Example 1 and 2 - 6PPD: 2.0 parts — Example 1 and 2 - Carbon black N330: 3.0 parts — Example 1 and 2 - S: 1.5 parts — Example 1 and 2 - CBS: 1.8 parts — Example 1 and 2 - Mixing temperature: 80 °C — Table 2 - Mixing temperature: 90 °C — Table 2 - Mixing temperature: 100 °C — Table 2 - Mixing temperature: 120 °C — Table 2 - Mixing temperature: 150-160 °C — Table 2 - Milling temperature: about 70 °C — Table 3 - Rheometry temperature: 160 °C — Table 4 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - low styrene content: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Rubber compounding / thermomechanical mixing; no SBR polymerization medium disclosed

low styrene content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Examples 1 and 2: Natural rubber was modified with unsaturated silanes in silica-filled formulations. - Example 1: Acryloxypropyltrimethoxysilane-based silane 2 was used. - Example 2: Comparative silane 1 was used for comparison. - Examples 3 and 4: Alternative silanes were evaluated for silica dispersion and rolling resistance. - Example 5: Bis(triethoxysilylpropyl)-tetrasulfane and related silanes were compared.

Technical Relevance This patent is only indirectly relevant to low-styrene SBR because it addresses silane coupling and silica-filled elastomer compounding rather than SBR synthesis. It is useful as a downstream formulation reference for silica interaction, but it does not disclose SBR preparation.

Patent 9

- Patent Number: EP2986647B1
- Patent Title: Oil extended functionalized styrene-butadiene copolymer
- Assignee: ARLANXEO Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses emulsion polymerization of styrene-butadiene latex, followed by blending with vegetable-oil emulsions and coagulation/washing to obtain oil-extended functionalized SBR. A functionalized latex embodiment is prepared by incorporating glycidyl methacrylate at the start of polymerization and then ring-opening the epoxy group. - Water: 8,700 g — Example 1 - Emulsifier batch with potassium salt of fatty and rosin acids: 2,330 g — Example 1 - 1,3-Butadiene: 3,830 g — Example 1 - Styrene: 1,486 g — Example 1 - Pinane hydroperoxide: 2.5 g — Example 1 - Activator batch: 390 g — Example 1 - t-Dodecyl mercaptan: 10 g — Example 1 - Temperature: 7 °C — Example 1 - Conversion: 60% — Example 1 - Duration: about 7 hours — Example 1 - Oil in dry rubber: 37.5 phr — Example 1 - Water: 8,500 g — Example 4 - Emulsifier batch with potassium salt of fatty and rosin acids: 2,330 g — Example 4 - Buffering system pH: 8.7 — Example 4 - 1,3-Butadiene: 3,830 g — Example 4 - Styrene: 1,242 g — Example 4 - GMA/styrene blend: 106/133 g — Example 4 - Pinane hydroperoxide: 2.5 g — Example 4 - Activator batch: 390 g — Example 4 - t-Dodecyl mercaptan: 10 g — Example 4 - Temperature: 7 °C — Example 4 - Conversion: 60% — Example 4 - Duration: about 5 hours — Example 4 - Final latex pH: 9.2 — Example 4 - Oil in dry rubber: 37.5 phr — Example 4 - Bound styrene content: 21.0 wt% — Example 6 - cis-1,4 content: 9.5 wt% — Example 6 - trans-1,4 content: 76.3 wt% — Example 6 - vinyl 1,2 content: 14.2 wt% — Example 6 - Mooney viscosity (ML 1+4, 100°C): 45.8 — Example 6 - Average molecular weight: 356,000 g/mol — Example 6 - Styrene content: 15 to 50 wt% — Claims - Butadiene content: 42 to 80 wt% — Claims - Functional monomer content: 5 to 43 wt% — Claims - Functional monomer content: 0.3 to 10 wt% — Claims - Mooney viscosity: 35 to 65 — Claims - Average molecular weight: 100,000 to 2,200,000 g/mol — Claims - Medium & Water Role: Water is the emulsion

polymerization medium and is also used later for latex/oil blending, coagulation, and washing of crumbs. The examples explicitly charge water to the reactor and wash the coagulated crumbs with demineralized water. - low styrene content: Bound styrene content: 21.0 wt% — Example 6

Medium & Water Role - Water is the emulsion polymerization medium and is also used later for latex/oil blending, coagulation, and washing of crumbs. The examples explicitly charge water to the reactor and wash the coagulated crumbs with demineralized water. - Core reaction medium: Aqueous emulsion / latex polymerization

low styrene content - Bound styrene content: 21.0 wt% — Example 6

Relevant Experimental Evidence - Example 1: Standard SBR latex was prepared by emulsion polymerization and extended with oil emulsion. - Example 4: GMA was introduced at the start of polymerization to make functionalized SBR latex. - Example 4: The final latex had pH 9.2 and was oil-extended with 37.5 phr oil. - Example 6: Functionalized SBR extended with epoxidized soybean oil showed bound styrene 21.0 wt% and Mw 356,000 g/mol.

Technical Relevance This patent is highly relevant because it is an emulsion SBR platform with explicit styrene/butadiene composition ranges and a functional monomer strategy that yields a low-to-moderate styrene functionalized SBR. It also demonstrates vegetable-oil extension and latex coagulation/washing, both important for practical SBR manufacture.

Patent 10

- Patent Number: US20170260340A1
- Patent Title: Styrene-butadiene rubber (sbr)-nanocarbon filled masterbatches and uses thereof
- Assignee: Amril Ag
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses preparation of nanocarbon-reinforced SBR masterbatches by dispersing carbon nanotubes in water with surfactant, combining the aqueous nanocarbon dispersion with SBR latex, and then coagulating the latex to recover a filled rubber masterbatch. It is a compounding and dispersion patent rather than an SBR synthesis patent. - Nanocarbon dispersion: 3% — Example 1.1 - Nanocarbon: 15 g — Example 1.1 - Surfactant: 75 g — Example 1.1 - Distilled water: 410 g — Example 1.1 - Stirring speed: 80 rpm — Example 1.1 - Stirring time: about 10 minutes — Example 1.1 - Ball milling time: 24 hours — Example 1.1 - pH of CNT dispersion: 8.0 to 10.0 — Example 1.1 - SBR latex concentration: 45% — Example 1.2 - SBR latex dilution: 30% — Example 1.2 - Surfactant level: about 5 pphr — Example 1.2 - 3 pphr CNT-SBR masterbatch — Example 1.2 - Mixing time: 60 minutes — Example 1.2 - Stirring speed: 80 to 100 rpm — Example 1.2 - Temperature: 23 to 30

°C — Example 1.2 - Calcium chloride: 500 ml of 20% w/w — Example 1.2 - CNT length: less than 50 µm — Examples - CNT outer diameter: less than 20 nm — Examples - CNT C-purity: greater than 85% — Examples - Medium & Water Role: Water is central to the nanocarbon dispersion and the SBR latex masterbatch preparation. The patent does not disclose SBR polymerization; instead, it uses water for dispersion, latex blending, and coagulation of the filled latex. - low styrene content: Not disclosed in extracted evidence

Medium & Water Role - Water is central to the nanocarbon dispersion and the SBR latex masterbatch preparation. The patent does not disclose SBR polymerization; instead, it uses water for dispersion, latex blending, and coagulation of the filled latex. - Core reaction medium: Aqueous latex blending / masterbatch preparation

low styrene content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1.1: A 3% CNT dispersion was prepared in water with surfactant and ball milling. - Example 1.2: CNT dispersion was combined with SBR latex to make a 3 pphr CNT-SBR masterbatch. - Example 1.2: The filled latex was coagulated with calcium chloride. - Example 2.2: CNT-filled SBR showed much lower electrical resistivity than virgin SBR. - Example 3.1: CNT addition increased Mooney viscosity, and mechanical peptizer reduced it.

Technical Relevance This patent is relevant to SBR processing and formulation, but not to SBR synthesis. It demonstrates how SBR latex can be combined with aqueous nanocarbon dispersions and coagulated, which is useful for downstream handling of SBR-based materials.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	low styrene content	Key Findings
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US8530017B2	Korea Kumho Petrochemical Co., Ltd.	2013	Water is the polymerization medium in the emulsion SBR synthesis, and later water is also used in downstream coagulation/stripping and washing steps. The examples explicitly charge water to the reactor and later describe stripping and drying after polymerization.	Styrene monomer amount: 10-50 wt % — Claims	Example 1: Reactive emulsifier-containing SBR was prepared with 260 g styrene and 710 g butadiene, giving Mw 860,000 g/m
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US6858683B2	Polimeri Europa S.P.A.	2005	The polymerization itself is carried out in cyclohexane solution, not in water. Water appears later only in the solvent-removal step, where the polymer solution is fed to a stirred reactor containing water heated by steam to eliminate solvent, followed by filtration and drying.	Styrene: 3 wt % — Comparative Example 2; Styrene: 3 wt % — Comparative Example 4	Comparative Example 1: Batch solution SBR was made in cyclohexane and finished by steam stripping and drying.
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US9328176B2	Korea Kumho Petrochemical Co., Ltd.	2016	Water is part of the emulsion polymerization medium and is also used later in aqueous ring-opening chemistry and downstream latex handling. The examples explicitly charge water to the reactor and use 20% aqueous sulfuric acid, NaOH, or KOH for post-polymerization ring opening.	Styrene monomer amount: 10-50 wt % — Synthesis passages	Example 1: GMA-containing SBR was prepared in water and ring-opened with 20% sulfuric acid to give hydroxyl-functional c
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US8455600B2	Korea Kumho Petrochemical Co., Ltd.	2013	Water is the polymerization medium for the SBR latex synthesis, and later water is also used in latex handling and compounding-related steps. The examples explicitly charge water to the reactor and describe the latex as the product form before oil blending and compounding.	Bound styrene content: 21.0 wt% — Example 6	Example 1: Reactive polyol was incorporated during emulsion polymerization to give a styrene-butadiene-reactive polyol c
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US2014025685A1	Robert Anthony Bethea	2014	The polymerization is carried out in hexane solution, not in water. Water appears later only in steam-stripping/solvent-removal operations used to recover the oil-extended SSBR composite.	Not disclosed in extracted evidence	Example I: A hexane solution with 60 wt% butadiene and 40 wt% styrene was anionically polymerized to high conversion.
US10174133B2	Lg Chem, Ltd.	2019	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Comparative Example: Comparative Example

EP2657262B1	The Goodyear Tire & Rubber Company	2014	The polymerization is conducted in hexane solution, while water is used later in steam stripping to remove solvent from the SSBR/oil blend. No aqueous polymerization medium is disclosed.	Not disclosed in extracted evidence	Example 1: Base SSBR was prepared from 70 wt% butadiene and 30 wt% styrene in hexane.
EP2424740B1	Dow Corning Corporation	2013	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Examples 1 and 2: Natural rubber was modified with unsaturated silanes in silica-filled formulations.

EP2986647B1	ARLANXEO Deutschland GmbH	2017	Water is the emulsion polymerization medium and is also used later for latex/oil blending, coagulation, and washing of crumbs. The examples explicitly charge water to the reactor and wash the coagulated crumbs with demineralized water.	Bound styrene content: 21.0 wt% — Example 6	Example 1: Standard SBR latex was prepared by emulsion polymerization and extended with oil emulsion.
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US20170260340A1	April 2017	2017	Water is central to the nanocarbon dispersion and the SBR latex masterbatch preparation. The patent does not disclose SBR polymerization; instead, it uses water for dispersion, latex blending, and coagulation of the filled latex.	Not disclosed in extracted evidence	Example 1.1: A 3% CNT dispersion was prepared in water with surfactant and ball milling.
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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Emulsion-route patents (US8530017B2, US9328176B2, US8455600B2, EP2986647B1) all use water as the polymerization medium and typically rely on rosin/fatty-acid emulsifiers, peroxide/redox initiators, mercaptan chain-transfer agents, and low temperatures around 7-10 °C.
- Solution-route patents (US6858683B2, US20140256858A1, EP2657262B1) use hydrocarbon solvents such as cyclohexane or hexane, organolithium initiation, and solvent stripping; water appears only in downstream stripping/workup.
- The explicitly low-styrene feed disclosure among the primary SBR synthesis patents is strongest in US6858683B2, where a continuous solution example uses 3 wt% styrene in the feed; EP2657262B1 uses 30 wt% styrene feed and US20140256858A1 uses 40 wt% styrene feed.

- Functionalization strategies differ: US8530017B2 uses reactive emulsifiers, US9328176B2 uses epoxy-acrylate monomer plus aqueous ring-opening, US8455600B2 uses reactive polyols, and EP2986647B1 uses GMA plus vegetable-oil extension.
- US20170260340A1 and EP2424740B1 are downstream compounding/masterbatch or silane-modification patents rather than SBR synthesis patents; they are relevant to processing and filler interaction but do not define low-styrene SBR synthesis conditions.

4. CONCLUSION

Across the selected primary patents, low-styrene SBR is realized mainly through emulsion or solution polymerization of styrene and butadiene with careful feed control, then enhanced by functional comonomers, reactive emulsifiers, or post-polymerization extension to recover performance lost by reducing styrene. The most explicit low-styrene feed disclosure appears in the solution-process patent US6858683B2, while the emulsion patents emphasize silica affinity and tire performance through functionalization rather than extreme styrene reduction. Water is central in the emulsion patents and appears only as a downstream stripping/workup medium in the solution patents.

5. REFERENCES

- US8530017B2 | ... copolymer using reactive emulsifier and styrene-butadiene copolymer prepared by ... | Korea Kumho Petrochemical Co., Ltd. | US | 2013 | <https://patents.google.com/patent/US8530017B2/en>
- US6858683B2 | Process for the preparation of SBR rubbers with an improved processability and ... | Polimeri Europa S.P.A. | US | 2005 | <https://patents.google.com/patent/US6858683B2/en>
- US9328176B2 | Functional styrene-butadiene copolymer | Korea Kumho Petrochemical Co., Ltd. | US | 2016 | <https://patents.google.com/patent/US9328176B2/en>
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- US20140256858A1 | Solution polymerization prepared styrene/butadiene elastomer containing liquid ... | Robert Anthony Bethea | US | 2014 | <https://patents.google.com/patent/US20140256858A1/en>
- US10174133B2 | Modified styrene-butadiene copolymer, preparation method thereof, and rubber ... | Lg Chem, Ltd. | US | 2019 | <https://patents.google.com/patent/US10174133B2/en>
- EP2657262B1 | Triglyceride containing solution polymerization prepared styrene/butadiene ... | The Goodyear Tire & Rubber Company | EP | 2014 | <https://patents.google.com/patent/EP2657262B1/en>
- EP2424740B1 | Elastomer compositions modified by silanes | Dow Corning Corporation | EP | 2013 | <https://patents.google.com/patent/EP2424740B1/en>

- EP2986647B1 | Oil extended functionalized styrene-butadiene copolymer | ARLANXEO Deutschland GmbH | EP | 2017 | <https://patents.google.com/patent/EP2986647B1/en>
- US20170260340A1 | Styrene-butadiene rubber (sbr)-nanocarbon filled masterbatches and uses thereof | Amril Ag | US | 2017 | <https://patents.google.com/patent/US20170260340A1/en>